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Original article

Water science in the public sphere: A narrative network analysis of fracking and oil spills in Colombian media

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ABSTRACT

The looming global water crisis and waning dependence on fossil fuels have intensified the public discourse on oil and water rights, leading to conflicts over the water impacts of oil extraction and transport. This article assessed water-oil controversies that dominated media coverage in Colombia between 2017 and 2021. By drawing on discourse analysis, the assessment exposed how debates are shifting away from an economic-environmental binary, towards nuanced discussions surrounding complexity, uncertainty and the role of expert knowledge. A database of 388 newspaper articles was compiled to characterize the competing coalitions, storylines and discourse frames for each controversy and a subset ($n = 168$) were systematically coded. The relative prominence of discourse frames and links among them were quantified and visualized using network analysis methods. The findings presented a structural understanding of each coalition's narrative network, providing insights regarding the relative success of each within the public sphere. Environmental narratives in Colombia have transcended traditional political and ideological barriers by emphasizing issues of democracy, scientific uncertainty, and the precautionary principle, thus moving beyond unilateral economic and environmental debates.

1. Introduction

On September 15, 2022, politicians, activists and community leaders gathered for a public audience before the Colombian Senate. This was a new kind of public audience, one in which environmental debates were supported by the new government, who has vowed to convert Colombia into a "global potency of life." A young senator wearing an anti-fracking t-shirt convened the meeting. She and others sported water bottles filled with mucky water, packaged with the logo *"Agua piloto: el sabor del futuro"* (Pilot water: the taste of the future). The brand had been designed to raise awareness about the threat hydraulic fracturing or "fracking" poses to the future of water and life. Influencers promoted the product on social media, and activists staged sales of "pilot water" on the streets, asking fracking supporters to try drinking the water of the future. Now, the bottles accompanied a public audience to discuss the proposal of the new Law 114 of 2022, which would prohibit fracking in Colombia and give the national government two years to formulate a coherent energy transition policy. Susana Muhamad, the recently appointed Minister of the Environment, opened the session. She praised the openness and democratic nature of the debate, inviting the public to contribute their views. During her intervention, she also stated:

> "What fracking does is to lock us in deep within an unsustainable energy matrix, leaving us with immense environmental uncertainties. Because we have no idea what fracturing the bedrock in Barrancabermeja would mean, due to its globally unique characteristics."

I suggest that Muhamad's emphasis on the uncertainty of environmental impacts and the complexity of Colombia's unique hydrogeology and ecosystems represent shifting views of water science that have become consolidated in the Colombian public sphere through water-oil controversies in recent years. Narratives that emphasize the unknown and the complex have become the opening points for increased public participation, supporting a wave of diverse expressions of environmental democratization and water justice movements across the country (Dietz, 2019; Roa-García, 2017).

Muhamad's main points were echoed in various forms throughout the public audience. Although those who spoke before the Senate represented heterogeneous interests, objectives, and priorities, this has not always been the case in Colombia, in matters of water and oil

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controversies. The Colombian public sphere has historically been divided between pro-economy and pro-environment fractions, heavily marked by the dynamics of internal conflict. Public debates about oil tended to become "commitment rituals" whereby participants either sided with large-scale capital or risked "being accused of collaborating with the guerrillas" (Tognato, 2017, p. 101). For decades, issues such as the national royalty system or the nationalization of the oil industry were divided along these lines. In recent years, the public debate on fracking has united actors across the ideological and political spectrum, leading to the formation of a broader coalition (Tognato, 2017). An analysis of the early stage of the fracking controversy in Colombia (2011 to 2018) found that the state's discourse echoed that of the oil industry, emphasizing the urgency of fracking for economic development, and systematically minimizing the public's concerns (Roa-Avendaño and Orduz-Salinas 2021). As this rhetoric intensified, demands for water justice through mechanisms of democratic participation were on the rise (Dietz, 2019; Roa-García, 2017).

This article seeks to comprehend the narrative and discursive elements that emerged from 2018 to 2020, that allowed Colombians to move beyond earlier "commitment rituals" in water-oil controversies. I ask, what elements of discourse surrounding water science in the public sphere have allowed for the creation of a broader coalition? How have matters of complexity, uncertainty, and the role of expert knowledge figured within the narratives of opposing coalitions? In asking these questions, I recognize that the issues at stake are not so black-and-white. Society cannot be understood as merely consisting of discrete opposing groups and controversies cannot be understood as merely comprised of objective facts. These categories are fluid and could better be understood on a spectrum: coalitions and the objective "facts" of controversies are constantly shifting. I do not pretend to capture these heterogeneities or nuanced understandings in a single article. Rather, I use the methodological lens of "discourse coalitions," which are considered as coherent groups of actors who use similar lines of argumentation to communicate their stance and are loosely held together by overarching narratives or "storylines" about specific issues. This concept of coalition is distinct from that developed by scholars of agrarian conflict in Colombia, who have studied coalition formation among indigenous, afro-descendant, and rural women's groups in enforcing redistributive land reforms (Coronado, 2019), as well as coalitions among state, elite and paramilitary actors engaged in rural land grabs (Grajales, 2017; Reina, 2022). By adopting the lens of "discourse coalitions," I acknowledge that a binary of grouping of actors that side with industry versus the environment is artificial, and coalitions are constantly in flux.

I accept this paradox for an analytical moment, to tease out how the public discourse on water science in Colombia is grappling with understandings of complexity, uncertainty, and the role of expert knowledge. It is also important to note that these discussions do not occur in a vacuum, and the public sphere is not only shaped by these notions. Behind each discourse framing or narrative discussed in this paper, there is a world of political economic factors shaping these views. I make brief mention of some of these factors influencing the shape of discourse in the following section. Each framing could be analyzed and understood within the context of long historical processes that has brought it to the fore. Therefore, I urge readers to view the results of this study as a useful analytical exercise - a contribution towards a larger narrative with many moving parts - without adopting a binary or stagnant view of water-oil controversies in the Colombian public sphere.

To answer these questions, I present a detailed and systematic analysis of two of the most prominent water-oil controversies occurring in Colombian public media during this timeframe: an oil spill that occurred in the Middle Magdalena region of Colombia in 2018 and the fracking debate as it played out from 2018 to 2020. I draw on argumentative discourse analysis and network analysis to quantify and visualize the ways in which discourse framings have addressed complexity, uncertainty, and the role of expert knowledge. I suggest that environmental narratives in Colombia have been successful in transcending traditional political and ideological barriers, by emphasizing issues of democracy, scientific uncertainty, and the precautionary principle, thus moving beyond unilateral economic and environmental debates. This shift can be understood as a disjuncture in the public's understanding of water science. Rather than relying on technoscientific expertise in either environmental or economic calculations of risk, a multiplicity of actors point to the irreducible nature of uncertainties and complexities (Funtowicz and Ravetz, 1994), opening up water science to actors involved in environmental democratization and water justice movements (Roa-García, 2017).

2. Water-oil controversies and public discourse

Water science and oil policy involve complex relationships among technoscientific and political economic factors. Truths and facts that are presented as objective, are often produced through the power relations of subterranean geopolitics (Valdivia, 2015) as well as the socio-material interactions of technical, chemical, and legal processes (Sawyer, 2015). In Colombia, the state's discourse on water and oil has traditionally sided with the oil industry, which has been a major provider of financially attractive foreign investments. For example, from around the 1980s to the 2000s, government narratives would emphasize the importance of fossil fuel production as a source of public income, used for improving public services and infrastructure. Towards the 2010s, the state narrative shifted towards promoting the extractive sector as a major financer of Colombia's Peace Accord (Strambo and González Espinosa, 2020). The oil industry has also been historically linked to the internal dynamics of armed conflict, with evidence of multinational corporations having financed the war system (Richani, 2005). Those who live in areas dedicated to oil production have suffered the social and environmental consequences and "slow violence" of ecosystem degradation, without the promised benefits to livelihoods (Vélez-Torres and Méndez, 2021). These dynamics have polarized narratives around oil, so that scientific "facts" presented as "neutral, objective resource[s] for accountable and transparent decision-making" have been shaped by multiple political economic interests (Nost and Goldstein, 2022, p. 4).

Public narratives on water-oil controversies are often narrow or simplistic, and journalists' reports are frequently shaped by the geographic, economic and political contexts of news production (Matthews and Hansen, 2018). In the case of fracking, studies have found that positions regularly vary based on demographics (Brasier et al., 2011; Jacquet, 2012) or media effects (Boudet et al., 2014; Cunningham, 2019; Habib and Hinojosa, 2016). For example, in the US, news coverage on fracking employed framings unique to the local political, economic and social context of each state (Hedding, 2017). In Canada, news stories on fracking differed by province and focused either on environmental issues or economic benefits, but rarely both (Olive, 2016). Scientific "facts" regarding water availability have been mobilized by national governments to garner support for the extractive industry. For example, in Argentina, the provincial government of Neuquén downplayed the complexity of water use and emphasized a perception of surface water abundance to gain public consent for fracking (Bernáldez and Herrera, 2019).

Uncertainty due to incomplete knowledge is a central aspect of water-oil controversies. This can stem from insufficient scientific evidence of risk, a lack of adequate data, or challenges with transparency. The public has challenged industry secrecy, for example, in the case of the chemical content of fracking fluids (Kinchy and Schaffer, 2018) or contradictory oil spill data (Watts and Zalik, 2020). Actors involved in water governance and fracking have often agreed on the need for more data regarding geophysical factors, while issues of responsibility and the incorporation of local and ancestral knowledge have remained unresolved (Moore et al., 2018). Uncertainty has been studied both as a weapon and as an opportunity. Scholars have explored how corporations and the state weaponize uncertainty, by strategically "deny[ing]

awareness of things it is not in their interest to acknowledge" (McGoey, 2016, p. 4). For example, Hesse (2021) demonstrated how the fracking industry capitalized on the spatial and temporal flexibility of worksites to render health risks unknowable and ungovernable. Alternatively, uncertainty can lead to pragmatic and improvised practices of survival (Cavallo, 2021; Martin, 2021; Senanayake, 2021), as well as opportunities for "a more rigorous, robust, transparent – and democratically accountable – environmental politics" (Stirling, 2018, p. 120).

Uncertainty in water-oil controversies can lead to power dynamics regarding the role of expert knowledge. In the US, researchers have found that government knowledge was privileged over all other categories of evidence in rule-making processes surrounding fracking regulations (Baka et al., 2019). In Alberta, Canada, newspapers portrayed the oil and gas industry as victims of ideologically motivated environmental lobbyists during oil sands protests (Gunster and Saurette, 2014). A similar narrative was identified in Colombia during popular consultations against extractive activities (Dietz, 2019). The state framed consultations as emotionally and ideologically charged and motivated by political interests or affiliations with left-wing guerrilla groups, thus discrediting mobilized publics (Hess and Sovacool, 2020).

Expertise can also be discredited, as in the case demonstrated by Duffy, 2021, where industry actors influenced the interpretation of science by framing certain problems as unrelated to fracking. The case refers to a public outreach campaign launched by the Independent Petroleum Association of America (IPAA) to demonstrate the health and safety benefits of fracking. IPAA's key discourse strategy was to narrow their definition of fracking to refer to the specific process of well stimulation and to exclude processes such as well construction and waste disposal. This way, the campaign report could include statements such as "Nearly 65 years of use, and not one case of groundwater contamination caused by hydraulic fracturing" (Duffy, 2021, p. 65). In some cases, coalitions moved away from an emphasis on expert knowledge to highlight the importance of democracy, participation, and social license. In the UK, a shift in anti-fracking discourse has moved towards issues of local power and democracy over those of environmental risk or economic benefits (Bomberg, 2017).

The intersection of water science and the politics of oil offer a unique arena for the study of how science plays out in the public sphere. Water-oil controversies in media debates demonstrate the ways in which diverse actors address the complexity of water-oil dynamics, the uncertainty brought on by incomplete knowledge, and the role that expert knowledge plays in shaping policy. However, questions of oil extraction and water often go beyond notions of science, to "involve the very definition of the 'things' that are at stake" (Blaser, 2013, p. 14). For the U'wa people of Casanare, Colombia, for example, oil is the "blood of the mother earth" (*ruiría, sangre de la madre tierra*) (Duarte 2016: 387). Defending the territory from oil extraction is a way to defend "complex webs of relations between humans and nonhumans" (Blaser, 2013). In focusing on discourse framings surrounding complexity, uncertainty, and the role of expert knowledge, I do not wish to disregard the conceptual richness of these "ontological conflicts." Indeed, exploring society's shifting understandings of science may lead to greater dialog on conceptualizing science *otherwise*. For the present study, I draw attention to these shifting notions of water science in the public sphere, even as the political economy of extractives shows signs of change. I hope that others will further trace these changes in discourse and connect them with the moving ontologies of water and oil in Colombia and beyond.

3. Methods

The objective of the assessment was to identify the most recent and prominent water-oil controversies in Colombia and to decipher how meanings surrounding complexity, uncertainty and the role of expert knowledge were structured and communicated during these controversies. To achieve this aim, water-oil controversies were first identified through an overview of recent (2017–2021) Colombian news coverage. Argumentative discourse analysis (Hajer, 1995) was employed to characterize the competing coalitions, storylines and discourse frames for each controversy. A subset ($n = 168$) of articles were systematically coded using discourse frames identified through the argumentative discourse analysis. These discourse frames were categorized based on whether they communicated meanings surrounding complexity, uncertainty, or the role of expert knowledge. To determine how these meanings were structured during water-oil controversies, the relative prominence of discourse frames and links among them were quantified and visualized using network analysis methods.

3.1. Data collection

Initial data collection involved a compilation of news sources related to oil and water, using the Center for Research and Popular Education (*Centro de Investigación y Educación Popular*, CINEP) media database.[1] Using the descriptor code related to the oil and mining industries (Ecopetrol, OXI, petroleum politics, mining, hydrocarbons) for August 2017 to August 2021 returned a total of 3223 articles. Articles were then filtered by those that mentioned water which reduced the search to 945 articles. After excluding those related to mining or exclusively financial aspects of the oil industry, these were reduced to 388 articles.

These articles were categorized thematically based on headlines to identify the most prominent water-oil controversies within the period 2017–2021 (Fig. 1). Although other topics, such as offshore drilling and illegal refineries also surfaced in this analysis, the Lizama oil spill and fracking had the largest number of articles. After a preliminary reading of articles related to these two water-oil controversies, main actors and coalitions were identified. Peaks in the number of published articles were selected by graphing the articles out over time and identifying three-month periods with increased article publication for major controversies (Fig. 1.a- 1.d). A sub-set of articles ($n = 168$) released during these peaks were selected for systematic, manual coding.

3.2. Narrative network analysis

Argumentative discourse analysis (Hajer, 1995) was used to examine how meaning is created through argumentative struggles in which two or more "discourse coalitions" form. Discourse coalitions are coherent groups of actors who use similar lines of argumentation to communicate their stance. They are loosely held together by overarching narratives or "storylines" about specific issues. Storylines promoted by coalitions are comprised of "frames", or coherent clusters of concepts, sentiments and sets of assumptions shared by coalition members. Frame concepts are ideologically and intrinsically compatible (Roth and Bourgine, 2005), mixing information and emotive appeal to reconstruct meaning (Bomberg, 2017). In the context of this paper, coalitions and storylines corresponded to major water-oil controversies identified in step 1 (data collection): the Lizama oil spill (Fig. 1.a) and the fracking debate in Colombia (Fig. 1.b-d). Each discourse coalition was related to a storyline comprised of a series of supporting discourse frames.

The discourse frames were identified through several iterative rounds of qualitative analysis. During the initial analysis, groups of 2–3 sentences that made up a coherent argument were tagged with "a word or short phrase that symbolically assign[ed] a summative, salient, essence-capturing, and/or evocative attribute" (Saldaña, 2015, p. 3). This first round of coding through the inductive approach, a bottom-up

[1] The CINEP media database offers systematized news coverage from 10 national and regional Colombian news sources. It included over 700,000 articles compiled over 42 years, on topics relating to the social sciences, including religion and conflict, politics, and governance, narcotrafficking, culture and society, and ecology and environment. The database includes 163 thematic descriptors within these broad categories.

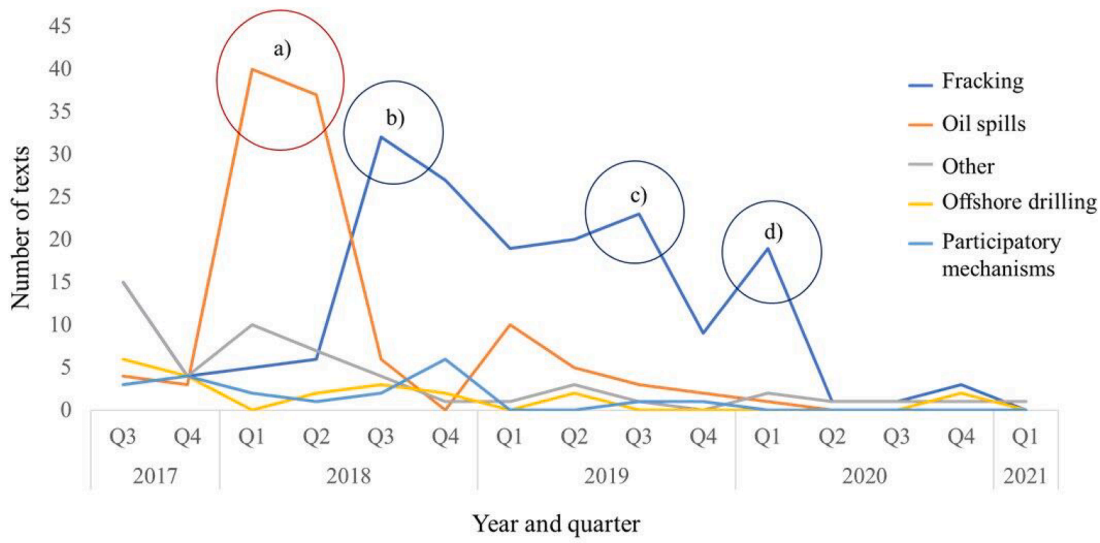

Fig. 1. Number of articles by water-oil knowledge controversy (August 2017 – January 2021). Selected data represent the following events:
a) Lizama oil spill
b) Court invokes precautionary principle to suspend norms on fracking
c) Expert recommendations on fracking made public
d) Pilot projects controversy.

approach that allows codes to emerge through interaction with the text. This allowed for a deeper appreciation of the complexity and diversity of the data and produced a long list of initial descriptive codes that were later consolidated into higher level analytical categories during subsequent rounds of coding (Skjott Linneberg and Korsgaard, 2019). The consolidation of initial codes was informed by discourse frames previously identified in the literature. Codes were then grouped by category to understand each coalition's narrative approach to complexity, uncertainty, and the role of expert knowledge.

Discourse analysis was combined with network analysis to explore the relative strength and relationships among discourse frames within three pre-established categories of concern (complexity, uncertainty and the role of the expert). Network analysis has been used in conjunction with discourse analysis to visualize actors, concepts, and storylines (Brugger and Henry, 2021; Hamann and Suckert, 2018).[2] Network graphs allowed for the visualization of nodes and links between nodes. In the context of this paper, nodes represented discourse frames and links represented the simultaneous use of two or more frames within a coherent argumentative unit. Nodes were sized based on the number of mentions, so that the discourse frames used most often by a particular coalition figured as larger within the network graph. This permitted a visual analysis of the most *prominent*, or most frequently used, discourse frames. Discourse frames that were often used together were represented by nodes with thicker links between them. Network graphs also placed more connected nodes towards the center of the graph, and less connected nodes towards the periphery. Discourse frames which were most *central* were most connected to other discourse frames. Finally, nodes were also color-coded by category, with gray nodes representing discourse frames related to complexity, yellow nodes representing discourse frames related to uncertainty, and orange nodes representing discourse frames related to the role of expert knowledge. Color-coding nodes granted a visualization of the *prominence* and *centrality* of nodes by category. This aided in the analysis of each coalition's approach to complexity, uncertainty, and the role of expert knowledge. Together, the placement, size and color of nodes and links within network graphs represented each coalition's *narrative network structure.*

4. Results

4.1. Major water-oil controversies

A preliminary reading of Colombian news articles on water-oil controversies (2017–2020) led to the identification of two major controversies: the Lizama oil spill of 2018 and the public debate on fracking (2018–2020). The following section offers a brief overview of these two controversies, drawing on media coverage of key events.

On March 2, 2018, an oil seep was registered 200 m from the Lizama 158 oil well in Barrancabermeja, Colombia. The well belonged to the state petroleum company, Ecopetrol, and had been closed in 2015, due to low production rates. At first, the oil spill went relatively unnoticed. It was not until March 15th that the full scope of the disaster was revealed to the public through media reports. Heavy rains exposed thousands of barrels of oil mixed with water and earth, contaminating a 24-kilometers stretch of the Sogamoso River and its tributaries. The oil spill lasted a month, causing the death and displacement of thousands of native animals, health effects in the local population, and economic consequences for local fishers. While Ecopetrol offered reassurance that the company had followed protocols, accusations against the company grew. A report from 2015 demonstrated that state authorities had warned Ecopetrol about the risk posed by the abandoned wells in the La Lizama oil field. For the month that followed the oil spill, controversies ensued about what caused the spill and the delayed response, as well as the extent of the damage and the expected duration of the recovery.

Throughout the controversy, several experts became involved in evaluating the causes and recovery efforts. Environmental toxicologists emphasized the extent and complexity of ecosystem-wide contamination, while other environmentalists and political scientists questioned Ecopetrol's contingency plans and lack of transparency. Still other

[2] The present study used Atlas.ti©, a computer-assisted qualitative data analysis software to code for discourse frames identified in the first phase of the analysis. Discourse frames were grouped by category of analysis (complexity, uncertainty, and the role of expert knowledge) for each discourse coalition. "Coherent argumentative units" of two to five consecutive sentences were often coded for more than one discourse frame. These units with overlapping discourse frames were calculated using co-occurrence tables, and cross-tabulated matrices were produced by Atlas.ti. Co-occurrence tables were then exported to create network graphs using the R project for statistical computing. Links in network graphs were weighted by degree of co-occurrence.

experts praised the amount of work the company had put in towards cleaning up the spill and compensating communities. Meanwhile, Ecopetrol had been promising its own expert knowledge that would clarify the underlying causes of the oil spill: a "snubbing unit[3]" from the United States had been sent to the site to collect data that would be analyzed by a team of national and international experts. Once the data had been collected, Ecopetrol officially sealed the well, and, in July 2018, turned in the final report. The report identified four main causes, thus officially closing the case.[4]

Although the fracking debate in Colombia had already been on the public radar for several years, it was not until 2018 that it gained momentum. Due to global interest in fracking in 2012, the General Comptroller of the Republic (*Contraloría General de la República*) had issued a concept to the Ministry of the Environment to consider the environmental impacts of fracking in any future decisions on licensing. By 2014, oil and gas companies were preparing to obtain licenses and operate. With nineteen identified blocks, the minister of Mining and Energy endorsed fracking within Colombia, confident that environmental reviews would follow and soon be approved. The process, however, was delayed, and companies were kept waiting for environmental permits. Meanwhile, criticism from civil society and activists grew stronger, both on social media and in the streets. A protest in San Martín, Cesar, in 2016, resulted in violent clashes with the state's Mobile Anti-Disturbance Squadron (Esmad). After this event, the social organization "Alliance for a Colombia Free of Fracking" (*Alianza Colombia Libre de Fracking*) was formed, consolidating a coalition of over 85 social and environmental groups and academic actors.

Presidential elections and the growing environmental movement intensified the public debate on fracking. In August of 2018, the department of Boyacá held a public audience against fracking that resulted in the suspension of seismic exploration activities by the Polish firm Geofisyka Torún. A popular consultation in October of 2018 sought to ban fracking at the municipal level in Fusagasugá and San Bernardo, two cities in the department of Cundinamarca, with 99% of voters opposed to fracking. The municipal-level popular consultation offered the environmental movement hope that the activity could be banned at the base, through popular education and grassroots democracy. This hope, however, was short-lived. The same month, a court decision declared popular consultations regarding extractive activities to be null, since, legally, the state was the sole owner of the subsoil.

Around the same time, thirty congressmen were proposing legislation to the State Council to ban fracking at the national level, and the Comptroller General was requesting that the state suspend all activities advancing the development of fracking. The proposed legislation, drafted by the *Alliance for a Colombia free of Fracking*, and the Comptroller's requests, were based on the precautionary principle,[5] which had been embedded in Colombia's 1991 state constitution. The precautionary principle has been an important ethical consideration in hydraulic fracturing due to its assertion that potentially harmful processes subject to scientific uncertainty should not be permitted until the social benefits can be proven to outweigh the risks. In the case of fracking, the precautionary principle has been used as a rationale for moratoriums at state and subnational levels in Argentina, Brazil, Bulgaria, Scotland, the UK, France, the Netherlands, Canada (Quebec) and the United States (the states of New York and Maryland), due to its potential threats to human health (Patterson and McLean, 2018; Pérez Castellón et al., 2016). In response to the political climate and demands of the oil and gas industry, president Iván Duque announced the formation of an "independent interdisciplinary expert commission"[6] in October of 2018. The commission was tasked to study the viability of fracking in Colombia, so that the state could base its decisions on a rigorous scientific debate. In early November, the State Council declared a temporary moratorium on fracking.

After six months of investigations, the "independent interdisciplinary expert commission" turned in its report to the president. The commission, comprised of thirteen experts, had not given a definitive green or red light to fracking. Instead, the report highlighted environmental, ecological, hydrological, atmospheric, seismic and health hazards within the Colombian context. The report drew attention to the information gaps surrounding biodiversity and groundwater. In addition, it identified a lack of institutional capacity to monitor and control the projects, as well as the absence of social license in historically vulnerable and impoverished extractive regions. As a result, the commission recommended that, once several social and institutional factors had been addressed, research pilot projects (PPII, *proyectos piloto de investigación integral*) could be advanced.

The commission's report was controversial in many circles. Environmentalists were divided between those who whole-heartedly sanctioned the report, and those who rejected the recommendation for pilot projects. Industry hoped to interpret the recommendations as a green light for pilot projects. The state was left unsatisfied with the report's findings, having hoped for a clearer green light. The State Council called for a second committee from the National University to present answers to seven unresolved questions regarding the technical possibilities of mitigating risks. This second call resulted in two documents produced by two separate groups of experts, each with a different conclusion.[7] Because the results were not consistent between the two reports, Ecopetrol, the Ministry of Mining and the National Agency of Hydrocarbons filed an objection due to "serious error,[8]" asking the State Council not to consider the findings. Towards 2020, the debate centered on the nature and purpose of pilot projects, their link to potential commercial projects,

[3] A snubbing unit is a hydraulically powered rig used for heavy well intervention in wells that are still under pressure.

[4] The four reported causes were the following: 1) A group of contractors had been hired in November 2017 to determine whether to abandon or reactivate the well. While they were working a pipe fell and ruptured the "blanking plug" which was supposed to keep oil from welling up to the surface. 2) The casing pipe was corroded and experienced mechanical integrity failure. 3) Geological faults in the region provided pathways for the fluids to surface. 4) The overburden pressure was abnormally high due to the rock formations in the area.

[5] The precautionary principle, included in article 15 of the Rio Declaration on Environment and Development states: "In order to protect the environment, the precautionary approach shall be widely applied by States according to their capabilities. Where there are threats of serious or irreversible damage, lack of full scientific certainty should not be used as a reason for postponing cost-effective measures to prevent environmental degradation". In Colombia, the precautionary principle gained constitutional status with the enactment of Law 99 of 1993.

[6] The "interdisciplinary commission" was heavily skewed towards sciences traditionally affiliated with petroleum engineering, being comprised of 3 engineers, 3 geologists, 3 economists, 1 medical doctor, 1 zoologist, 1 lawyer and 1 philosopher. In addition, the Alliance for a Colombia Free of Fracking had denounced the fact that two members of the coalition (a geologist and petroleum engineer) had publicly announced that the country could prepare itself for fracking, one engineer had been the ex-director of the National Agency for Hydrocarbons and had worked for ExxonMobil, another had received research funding from Parex Resources, a company with a fracking contract, and that the lawyer had participated in regulatory adjustments to foment fracking in the state of Colorado.

[7] The first "expert response," a 460-page document written by six experts (political scientists, a chemist, physician, geologist and civil engineer) concluded that the risks could not be mitigated, while the second, a 177-page document written by two petroleum engineers and a geologist, concluded that they could be mitigated.

[8] An objection to an expert report due to "serious error" (*objeción del dictamen pericial por error grave*) is a legal objection that can be made when there is a *"difference between the reality of the object of study and the mental representation that the expert makes of it"* (Section 3, Sentence 5, March 2008, State Council, C. P. Enrique Gil Botero). It is not the same as an objection to expert evidence due to a lack of justification.

along with details for their licensing and monitoring.

4.2. Discourse coalitions, storylines, and discourse frames

Two opposing discourse coalitions were identified for each controversy (oil spills and fracking) .[9] For the oil spill controversy, the "industry defenders" coalition (comprised of industry representatives and networks, experts, research institutions and journalists) presented a storyline of the Lizama oil spill as a well-handled exception to construct meaning in ways that deflected the blame of the accident off the oil industry. The "industry accusers" coalition (comprised of residents of affected areas, local government officials, state authorities, experts, environmental NGOs and journalists) presented a storyline of the Lizama oil spill as a result of negligence to construct meaning in ways that placed the blame of the accident directly on the oil industry.

During the fracking controversy, the "pro-fracking" coalition included many of the same actors as the "industry defenders" coalition. The main storyline offered by this coalition was that of fracking as savior to construct meaning in ways that promoted fracking as a national economic strategy. The "precautionary" coalition included a broader coalition with NGOs (several under the umbrella of the *Alliance for a Colombia Free of Fracking* and the *National Environmental Movement*), academics, politicians, and journalists. The main storyline offered by this coalition was that of fracking as a threat to construct meaning in ways that would ban fracking activities from the national territory.

The discourse frames used by each coalition were identified and combined into frames used by pro-oil coalitions ("industry defenders" and "pro-fracking" coalitions) and pro-environment coalitions ("industry accusers" and "precautionary" coalitions). Discourse frames were grouped by categories related to complexity, uncertainty, and the role of expert knowledge (Fig. 2[10]).

Within the complexity category, several discourse frames addressed the issue of complexity directly. The *anomaly* frame addressed the surprising and unknowable events characteristic of complex systems. *Hydrogeological complexity* referred to Colombia's complex geology, groundwater, and landscape interactions, while *ecological complexity* referred to Colombia's complex ecosystems and high level of biodiversity. Other discourse frames within the complexity category indicated whether coalitions considered institutional, economic, political, environmental, and social concerns as isolated or interconnected problems. The *strong institutions* frame highlighted institutional experience and sound protocols, whereas the *weak institutions* frame emphasized corruption, oversight, and incompetence. The *security, opportunity,* and *development* frames described economic concerns. The *security* frame drew on arguments surrounding financial and energy security, the *opportunity* frame referred to untapped economic potential, and the *development* frame drew attention to the bounties of the national royalty system and an improved standard of living tied to the oil development model. The *alternative development* frame was a direct contradiction to the development frame, highlighting alternative visions of wealth and well-being. The *energy transition* frame presented fossil fuels as necessary for easing the gradual transition to renewable energies without harming the economy. On the other hand, the *new energy paradigm* frame called for an increased urgency to abandon fossil fuels and to question the growth paradigm in the face of the current climate crisis. The *social license and democracy* frame drew attention to issues of environmental democracy, public participation, and territorial autonomy. The *low risk* frame denied or minimized risks presented in the *environmental and health risks* frame. And the *geopolitics* frame referred to studies, policies and postures adopted by other countries, held up as standards for Colombia to follow.

Within the uncertainty category, knowledge was either presented as complete (*sufficient knowledge* frame) or incomplete (*insufficient knowledge* frame). Truths and facts presented by the state and industry could either be trusted (*transparency* frame) or met with skepticism (*non-transparency* frame). Areas of uncertainty were either dealt with through technoscientific protocols (*technical solutions* frame) or by exercising restraint (*precautionary principle* frame).

With regards to the role of expert knowledge, this was either invoked to bolster awareness of environmental and health risks (*expert evidence of risk* frame) or to confidently negate or minimize risk (*expert reassurance* frame). Perspectives regarding the public were also related to notions of expertise. The *lived experience* frame emphasized the value of knowledge gained empirically through daily life, whereas the *misinformed public* frame presented the public as unknowledgeable victims of false information. The *non-experts* frame referred to the disqualification of expertise on either side of the equation, whether the delegitimized experts were environmentalists, state authorities, or industry-sided professionals.

4.3. Narrative network structure

4.3.1. Oil spill as well-handled exception

The *oil spill as well-handled exception* storyline featured the following discourse frames as most prominent: *strong institutions, expert reassurance, technical solutions,* and *anomaly* (Fig. 3.a). Of these, the *technical solutions* frame was most central, or most connected to other discourse frames used by the coalition to support its storyline. This relationship suggests that the perception of institutional strength was promoted by dealing with uncertainty through technical solutions. As indicated by the width of links, the *strong institutions* frame was most often invoked along with the *transparency, anomaly,* and *expert reassurance* frames. The *misinformed public, sufficient knowledge,* and *low risk* frames were peripheral to the debate.

The *technical solutions frame* included statements that proposed handling or banishing uncertainty through the use of technology. The main object of uncertainty was the cause of the oil spill, although other questions arose regarding the delayed response, the likelihood of oil spills in other abandoned wells, the extent of the impact, and the length of the recovery. Ecopetrol's president Felipe Bayón repeatedly stated that nothing could be said regarding the causes of the oil spill until the "snubbing unit" from Houston, Texas had taken its measurements and a technical report had been delivered through these results.

The *anomaly frame* painted the Lizama oil spill as an exceptional event, one that had never occurred before in Colombia, and would probably never occur again. This view was also related to ideas that forces of nature, outside the company's control, contributed to the disaster. The coalition repeatedly mentioned heavy rains and possible seismic activity. Ecopetrol's final report on the causes of the oil spill relied on the *anomaly frame* to avoid responsibility. The report detailed four factors, the confluence of which qualified the event as "atypical, unpredictable, extraordinary and with almost inexistent probabilities of being repeated" (Aguirre Fernández, El Colombiano, 21 July 2018).

The *transparency* frame was linked to the *anomaly* frame, since Ecopetrol emphasized that the company had the same amount of information regarding the causes of the oil spill as everyone else. However, about a month into the disaster, the Attorney General revealed that its office's 2016 report had warned Ecopetrol about the improper abandonment of a series of wells in the Lizama oil field. In the face of accusations and uncertainty surrounding possible negligence, Ecopetrol reaffirmed both its innocence and its commitment to transparency.

Finally, the *expert reassurance* frame was strengthened by citing expert visits supporting the storyline that the oil spill had been a well-handled exception. For example, a commission from the United Nations was sent to the site for two weeks and concluded that the spill was "medium-impact," guaranteeing ecological recovery within a year. Experts from the Alexander von Humboldt Biological Resources Research Institute also visited the area and "assured that the cleaning work was

[9] See appendix (Table A1) for a list of coalition member types and examples.
[10] For example quotes of each discourse frame, see appendix (Table A2).

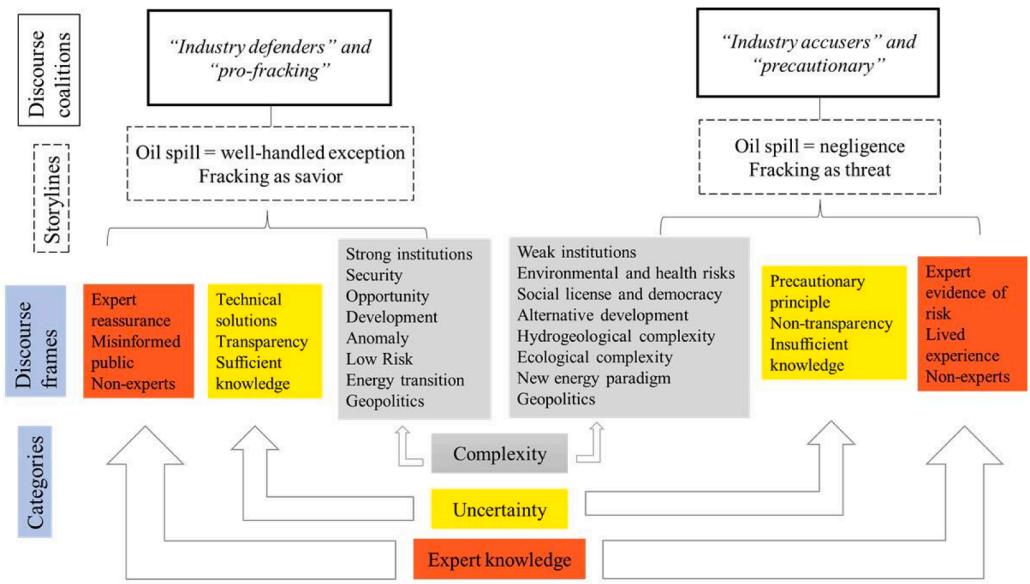

Fig. 2. Schematic of discourse coalitions, storylines, discourse frames and categories used for narrative network analysis of water-oil controversies in Colombia (2017–2021).

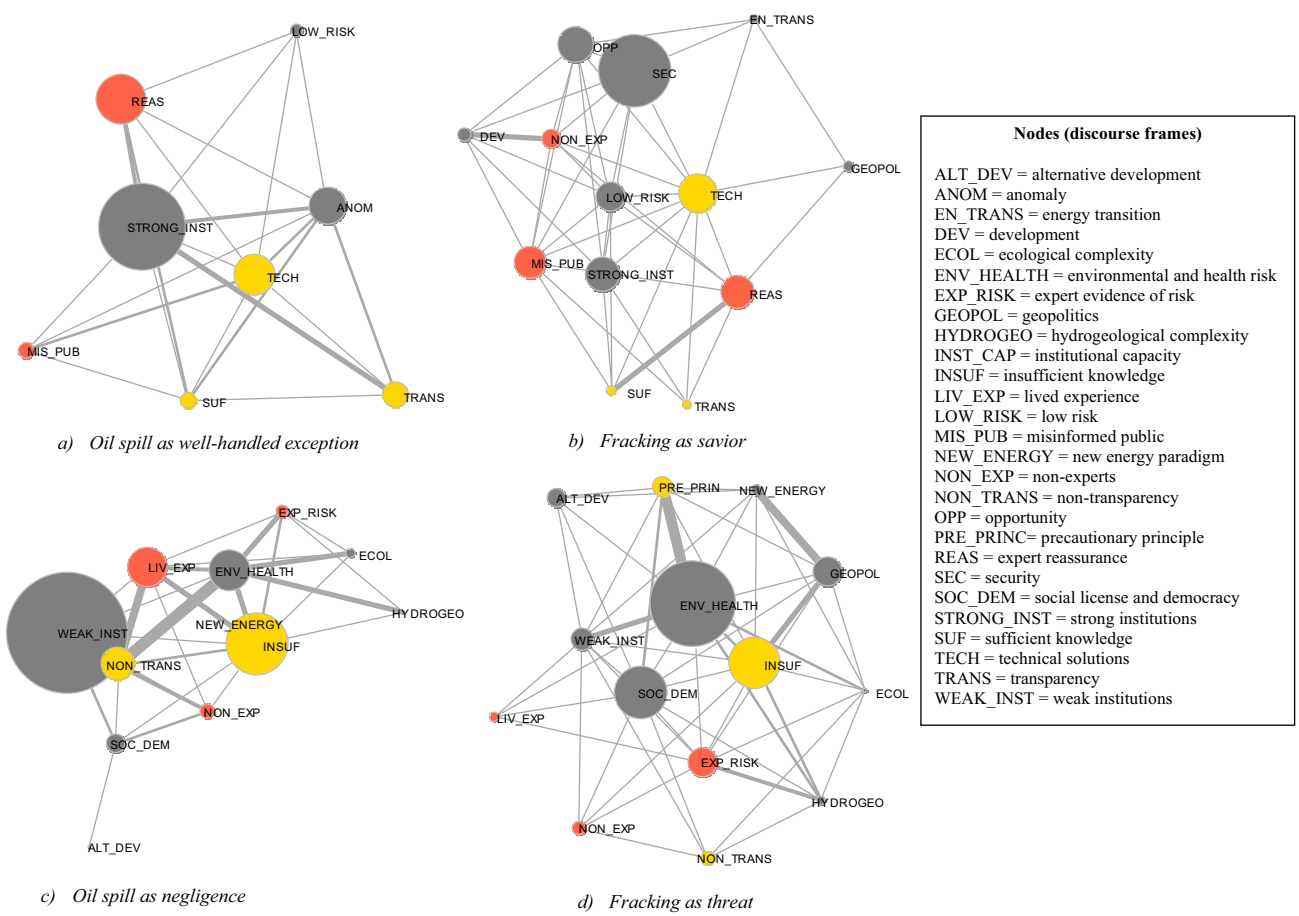

Fig. 3. Narrative network graphs of major water-oil controversies. Storyline a) Oil spill as well-handled exception was promoted by the "industry defenders" coalition. Storyline b) Fracking as savior was promoted by the "pro-fracking" coalition. Storyline c) Oil spill as negligence was promoted by the "industry accusers" coalition. Storyline d) Fracking as threat was promoted by the "precautionary" coalition. Node size represents the number of mentions of each discourse frame, while link width represents the number of co-occurrences (simultaneous mentions of two or more discourse frames within the same coherent argumentative unit). Nodes are color-coded by category (gray nodes are related to complexity, orange codes are related to expert knowledge, and yellow codes are related to uncertainty).

advancing satisfactorily" since "98% of the 1600 rescued animals… had been released into the Sabana de Torres nature reserve" (Vanguardia Liberal, April 6, 2018).

4.3.2. Fracking as savior

The *fracking as savior* storyline featured the *security* frame, together with the *opportunity* frame, as most prominent (Fig. 3.b). The *strong institutions* and *low risk* frames were secondary to frames related to economic benefits. With regards to the role of expert knowledge, the *expert reassurance, misinformed public* and *non-expert* frames were also relatively prominent, and most strongly linked to the *sufficient knowledge* frame. The *energy transition, development, sufficient knowledge, transparency,* and *geopolitics* frames appeared on the perimeter of the "pro-fracking" coalition's narrative network structure.

The *security* frame was related to both the idea of energy security as well as financial security and was used most by government and industry representatives. A common feature of the discourse around security was its emphasis on self-sufficiency, and the subsequent threat of having to decrease investment and increase fuel imports, deepening a national dependence on foreign powers. The *security* frame was closely related to the *opportunity* frame, which referred to the untapped potential and economic opportunities related to fracking. Both the *security* and *opportunity* frames were represented on the periphery of the network graph, indicating that although they were prominent frames, they were not often mentioned together with other discourse frames.

The *technical solutions* frame was the most central, along with the *low-risk* frame, allowing both to bring together all other discourse frames. The *technical solutions* frame was employed to point out technologies that would minimize environmental and health impacts. For example, coalition members stated that water recycling and wastewater treatment would eliminate the need to "sacrifice water for oil" (El Tiempo, March 8, 2020). The president of the Colombian Association of Petroleum (ACP) indicated that conventional oil extraction held similar risks to fracking, and therefore, there were already technical procedures in place to identify and mitigate risks. Later in the debate, once the expert commission had recommended pilot research projects, the "pro-fracking" coalition viewed these as a technical solution that would allow the government to "close existing uncertainties… and reach conclusions that would help make a decision" (Suárez quoted in El Tiempo, January 12, 2020).

The "pro-fracking" coalition used the *expert reassurance* frame to garner support for the *fracking as savior* storyline. When the president of Colombia, Iván Duque, announced the formation of the independent interdisciplinary expert commission, the "pro-fracking" coalition supported this development. Some members of the coalition conveyed their certainty that the experts' report would confirm the benefits of fracking, negate environmental and health risks, and reassure the public through sound, objective, quantifiable evidence. The Minister of Mining and Energy expressed her confidence in the expert commission, stating that "science and knowledge are the best way to clear the doubts of the Colombian people" (Suárez quoted in El Tiempo, March 8, 2020).

Together with the *expert reassurance frame*, the "pro-fracking" coalition also put forth the *misinformed public* frame. Regarding the public's misconception of fracking, Francisco José Lloreda stated that "many Colombians… have heard that this practice will be the end of our country's water resources and that they have to choose between water and fracking [but] this dilemma does not exist" (Lloreda quoted in El País, September 2, 2018). Coalition members also called on the need for the public to be educated "because no one has explained to them what fracking is about or the benefits it could bring" (Yanovich, El Espectador, March 10, 2020).

The *misinformed public* frame was supplemented by the *non-expert* frame, which discredited expert knowledge that opposed the main storyline. This frame was particularly apparent when the second expert committee, comprised of faculty from the National University of Colombia, presented their results before congress. The Ministry of Mining and Energy questioned the validity of the first set of experts from the National University before the court, citing "grave error" in their analyses. Finally, the *public bias* frame critiqued not only ignorance, but the manipulation of public opinion along ideological and political lines. The fracking debate was called "emotive," a "visceral and charged discussion" (Sánchez, El Nuevo Siglo, September 2, 2018) fueled by a Left that "want[s] to continue being poor" (Martínez, El Colombiano, September 13, 2019).

4.3.3. Oil spill as negligence

The *oil spill as negligence* storyline featured the *weak institutions* frame as most prominent, although the *environmental and health risk* frame was most connected, and the *insufficient knowledge* frame was both central and prominent (Fig. 3.c), suggesting that uncertainty was a central concern. The *weak institutions* frame highlighted how industry and environmental authorities were unprepared to handle surprises characteristic of complex systems. *Lived experience* was the most prominent frame with regards to expert knowledge. This frame referred to anecdotes about living with contamination (e.g., statements about residents' loss of fishing livelihoods, fish tasting like oil, lack of clean water, etc.) as evidence of the oil spill's extent. Compared to this frame, *expert evidence of risk* and *non-expert* frames appeared as secondary. *Lived experience* was also strongly related to *non-transparency* and *insufficient knowledge*.

The *weak institutions* frame highlighted flaws in emergency response plans due to a general lack of institutional capacity. One of the main concerns with the oil spill was the company's delayed response. Coalition members argued that "oil companies fail to maintain their infrastructure until an accident occurs" and that institutional capacity lags in isolated territories "*far from civilization*" (Velásquez quoted in El Colombiano, July 1, 2018). The coalition capitalized on the poorly handled incident to question the industry's ability to "be technically informed and able to respond to more advanced mechanisms, such as fracking" (Sánchez, El Colombiano, March 26, 2018).

Although not as prominent, the *hydrogeological complexity* frame stressed the importance of water resource impacts in oil regions with rich and complex groundwater systems. Ecopetrol president Felipe Bayón claimed that seismic activity could have induced the oil spill, since the region is characterized by high seismicity. Members of the "industry accusers" coalition turned his statements around, arguing that, precisely due to high levels of seismicity, more caution should be taken to ensure the safety of groundwater resources. The incident also drew attention to the issue of abandoned oil wells. Coalition members questioned the safety of abandonment practices in regions with complex groundwater systems. In the aftermath of the Lizama oil spill, one columnist wrote "whatever is uncovered about this case should serve as a warning bell regarding potential risks in other sealed and abandoned oil wells, of which there are hundreds across the expanse of the [national] territory" (quoted in El Nuevo Siglo, April 2, 2018).

4.3.4. Fracking as threat

The *fracking as threat* storyline featured the *environmental and health risks* frame as most prominent, although it was closely supported by two additional prominent and central frames: *insufficient knowledge* and *social license and democracy* (Fig. 3.d). The *environmental and health risks* was also most commonly invoked together with the *precautionary principle* frame, as well as the *weak institutions* frame. The *insufficient knowledge* frame was also linked to the *expert evidence of risk* frame, which was most prominent among frames dealing with the role of expert knowledge. The *new energy paradigm, alternative development, lived experience, non-expert, transparency, hydrogeological complexity,* and *ecological complexity* frames all figured within the debate, and although they were mentioned less, they were related to other frames and helped create a coherent narrative.

The *insufficient knowledge* frame emphasized uncertainty regarding the implementation of fracking within the Colombian "institutional and

biophysical context" (Ruiz quoted in El Tiempo, January 14, 2020). The argument surrounding "large information gaps about certain critical topics" (Rodríguez Becerra, El Tiempo, September 10, 2018) led to the formation of the expert commission. The *insufficient knowledge* frame was also used to support the *precautionary principle* frame, which "establishes that when there are indications of environmental harm, even if there is not complete scientific certainty, it is imperative to take all actions that the case requires to avoid these" (Rodríguez Becerra, El Tiempo, September 10, 2018).

The *social license and democracy* frame was pushed forward particularly through social movements, mobilizations, petitions, and statements made by the *Alliance for a Colombia Free of Fracking*. Early in the debate, participatory mechanisms at the departmental level (Boyacá) and municipal level (Fusagasugá and San Bernando) sought to determine their own autonomy and ban fracking from these territories. During this time, popular consultations were constitutionally binding. The governor of Boyacá, Carlos Andrés Amaya, who had campaigned for his department to be the first to ban fracking, said in one interview: "even if you don't like it, we are using a constitutional mechanism to defend our rights and territories" (quoted in El Espectador, September 3, 2018).

However, once popular consultations on extractive activities were suspended by the state in 2018, the "precautionary" coalition had to shift to other strategies to push for social license and democracy. The *Alliance* worked on obtaining signatures and drafting bills for congress instead. Due to these efforts, Óscar Sampoyo, a member of the alliance, called the court decision for a temporary moratorium a "triumph of citizen mobilization, of communities, workers, and a lot of people for the defense of water and life" (quoted in El País, September 11, 2019). The *social license and democracy frame* grew within the public debate on fracking with the recommendations made by the expert commission. The commission highlighted the need for obtaining social license from communities in oil regions before the implementation of fracking projects. Pilot projects would also have to "foment citizen participation" in all phases of planning and implementation (El Nuevo Siglo, September 4, 2019).

5. Discussion

The present study suggests that the widening of the "precautionary" coalition is related to its narrative network structure: its ability to embrace complexity, and to connect environmental and health concerns with issues of insufficient knowledge, democracy, and the precautionary principle. The ability to embrace complexity was observed through both the narrative network structure as well as direct references to characteristics of complex systems. In the field of political psychology, complex discourses have been defined as those capable of differentiation (considering distinct dimensions of a problem) and integration (making complex connections among differentiated topics) (Tetlock, 1993). This framework has been applied to measure the complexity of public discourse (Hoffman and Slater, 2007). In the context of this paper, the pro-environmental coalitions were able to use a variety of interrelated discourse frames to support their storylines, whereas the pro-oil coalitions tended to simplify water-oil controversies by focusing on fewer, less-connected economic concerns (i.e., the *security* and *opportunity* frames).

The tendency to disregard complexity led pro-oil coalitions to become contradictory. Using the *anomaly* frame, industry representatives claimed that most water-oil systems were not complex and did not present unanticipated events. For example, former Minister of Mines Ruben Darío Lizarralde stated that there was no oversight in Ecopetrol's operations, just "old wells that give surprises" (Lizarralde quoted in El Colombiano, 21 July 2018). Ecopetrol's president also sought to attribute the cause of the oil spill to possible seismic activity from one of the region's many geologic faults. This argument also presented the oil spill as an "unintended consequence of well-intentioned choices," a common response to accidents that "absolves technology and its human progenitors of responsibility and blame" (Jasanoff, 2004, p. 3).

Although the *anomaly* frame was applied to avoid liability, pro-environmental coalitions highlighted the frame's contradictory nature, using the same arguments to expose the complexity of the Colombian landscape. This was captured in both the *hydro-geological* and *ecological complexity* frames, which also figured during the fracking controversy. An emphasis on Colombia's diverse and "complex environment… its relief and geology, with great fluctuations between dry and wet seasons" (Carrizosa Umaña, El Espectador, September 4, 2018) led the Minister of the Environment to state that blanket risk assessments would not work. Instead, they would have to be conducted "point by point, village by village, aquifer by aquifer, community by community" (Ricardo Lozano, quoted in El Nuevo Siglo, August 15, 2018).

Pro-environmental coalitions were also successful in connecting environmental and health concerns with issues of uncertainty. The "pro-fracking" coalition tended to handle uncertainty with technical solutions. However, the "precautionary" coalition placed the *insufficient knowledge* frame at the center of its narrative, which was instrumental for the legal use of the precautionary principle. But it also allowed a more expansive view of expertise and public participation. In the case of the Lizama oil spill controversy, lived experience became important because scientific evidence was lacking, and transparency was a concern. Therefore, only residents could assess on-the-ground conditions. The *lived experience* frame combined contextual evidence with emotional and sensory ways of knowing. This is in line with recent scholarship regarding water security, which explores the dimension of water affect: "the emotional, cultural, and subjective experiences of water" (Jepson 2014, p. 109). Scholarship on toxicity has also focused on embodied and emotional suffering from water contamination (Perreault 2018), the experience of "slow violence" (Vélez-Torres and Méndez, 2021), and changes in connections to place due to water contamination (Willow, 2016).

An increased emphasis on uncertainty and lived experience during the oil spill controversy contributed to the prominence of the *social license and democracy* frame during the fracking controversy. The link between non-technical expertise, lived experience and democracy is a topic which merits additional attention, but is beyond the scope of this study. Scholarship on extractive knowledge controversies have found that expertise is often "a performative, ideological, and interactional phenomenon… authorized by existing power relationships" (Carmona and Jaramillo, 2020). Opening up expertise and including the non-technical, embodied experience of living with the impacts of oil extraction on water could lead to more democratic debates.

The "precautionary" coalition did not, however, focus on democratic principles in isolation. These were often presented together with the *expert evidence of risk* frame. Foreign expertise represented by the *geopolitical* frame also gained relevance, in the form of scientific studies and progressive policies. The coalition thus used simultaneous arguments regarding the need for social license, as well as scientific evidence of risks reported in other countries. This phenomenon of combining activist concerns with expert scientific evidence has been labeled "scientific environmentalism" (Yearley, 2005). Activists have moved environmental concerns into public discourse by drawing on expert authority (Pellizzoni, 2011) and increasingly applying scientific methods and languages (McCormick, 2007). Kimura (2016) analyzed this tendency in citizen science approaches, and warned against framing environmental problems through a reductive, technocratic lens, since doing so could unintentionally delegitimize actors less comfortable with the methods and language of science. Pro-environmental coalitions would do well to continue strengthening the discourse on democracy. An over-emphasis on scientific evidence could inadvertently create contradictions within the complex argumentative structure that has been built around uncertainty, democracy, and the precautionary principle.

5.1. Limitations

Although studies that analyze media sources can be rich and insightful, limitations arise due to several challenges with the press, including media capture and clientelism, journalist self-censorship, the silencing of Afro and indigenous voices, and a lack of coverage in remote regions – all of which result in untold, incomplete, and obscured stories. In the case of Colombian media outlets, most are owned by an elite group of politicians and media businessmen (Márquez-Ramírez and Manuel Alejandro, 2017). For example, *El Tiempo*, Colombia's biggest national newspaper group, was owned for a long time by the influential Santos family, who produced two of Colombia's presidents. Since 2012, *El Tiempo* has been owned by the banker Luis Carlos Sarmiento, the richest man in Colombia. Colombian journalists have been fired, censored, or threatened for exposing corruption and human rights abuses (Hughes et al., 2017). As a result, many have engaged in self-censorship, choosing certain stories over others to defend their livelihoods (Barrios and Miller, 2021). Through 16 in-depth interviews with Colombian journalists, Barrios and Miller found that "generalized fear, the distribution of official advertising and political publicity, the media's property, and local journalists' vulnerability" were the most prevalent motivations for self-censorship (1423). Some have responded by creating new independent online outlets, such as the *Silla Vacía*, a website founded in 2009. *Silla Vacía* has published articles on the fracking debate, and just recently on September 18, 2022, released its first documentary titled "Wells: a people marked by fracking" *(Pozos: un pueblo marcado por el fracking)*. Because this study used the CINEP database for its media analysis of oil spill and fracking debates in Colombia, results were confined to print sources. Online spaces such as the *Silla Vacía*, as well as discussions held on Twitter and other social media, were omitted. Other spaces where these controversies played out, including congress hearings, forums, academic conferences, street performances, and protests, were also not included. The scope of the analysis presented in this article was therefore limited and offers only a partial window into the structure of public discourse.

6. Conclusions

This article has highlighted two of the most debated recent water-oil controversies in Colombia: the Lizama oil spill of 2018 and the question of hydraulic fracturing or "fracking". It has detailed the coalitions, storylines and discursive frames formed around each controversy, both in support and opposition of the oil industry. Beyond the identification of discourse frames, the analysis has paid special attention to the ways in which these are structured and related to each other around themes of complexity, uncertainty, and the role of expert knowledge. An overview of each coalition's narrative network structure suggested that the precautionary coalition has achieved the greatest level of coherence in its discourse. Its success in the public sphere resulted from its placement of issues of scientific uncertainty and democratic participation at the core of its discourse, along with discussions of environmental and health risks. This allowed the coalition to transcend traditional political and ideological barriers and to move beyond unilateral economic and environmental debates. This shift can be understood as a disjuncture: a public break with technoscientific expertise in either environmental or economic calculations of risk. A multiplicity of actors point to the irreducible nature of uncertainties and complexities (Funtowicz and Ravetz, 1994) in water science, supporting movements for environmental democratization and water justice (Roa-García, 2017).

The analysis was conducted before the recent presidential elections in Colombia (2022) and therefore reflected the shape of discourse primarily during the Duque administration. Now that the state discourse on fracking is changing – the new Minister of the Environment and Minister of Mines (both women) oppose fracking, and the President is advocating for an energy transition – narratives on oil and water will assuredly shift. Already, there is a sense of this. In her presentation before the Senate on September 18, 2022, Muhamad emphasized the need to move beyond a "carbon culture" (*carbono-cultura*) in Colombia. Opening a discussion on culture and fossil fuel dependence could shift the public gaze away from the complexities and uncertainties of science to a discussion on the meanings of water and oil for culture, society, and the national imaginary. The present moment, and the state's commitment to leadership for an energy transition, offers new opportunities for exploring the shifting contours of these debates.

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